

COURSE OUTCOME 3

Date: 14-10-2025

Apply the concepts of modules and packages to design built in and user defined packages.

1. Create a package graphics with modules rectangle, circle and sub-package 3D-graphics with modules cuboid and sphere. Include methods to find area and perimeter of respective figures in each module. Write a program that finds area and perimeter of figures by different importing statements.
Surface area of cuboid : $2 * (\text{length} * \text{breadth} + \text{breadth} * \text{height} + \text{height} * \text{length})$
Volume of cuboid : $\text{length} * \text{breadth} * \text{height}$
Surface area of sphere : $4 * \pi * \text{radius}^2$
Volume of sphere : $(4/3) * \text{math.pi} * \text{radius}^3$

PROGRAM :

Graphics / rectangle.py

```
import math

def area(length, breadth):
    return length * breadth

def perimeter(length, breadth):
    return 2 * (length + breadth)
```

Graphics / circle.py

```
import math

def area(radius):
    return math.pi * radius * radius

def perimeter(radius):
    return 2 * math.pi * radius
```

ThreeDgraphics / cuboid.py

```
import math

def Surface_area(length, breadth, height):

    return 2 * (length * breadth + breadth * height + height * length)


def Volume(length, breadth, height):

    return length * breadth * height
```

ThreeDgraphics / sphere.py

```
import math

def surface_area(radius):

    return 4 * math.pi * radius * radius


def volume(radius):

    return (4/3) * math.pi * radius**3
```

Main.py

```
from Graphics import rectangle, circle
from Graphics.ThreeDgraphics import cuboid, sphere

# Rectangle
length = int(input("Enter length of rectangle: "))
breadth = int(input("Enter breadth of rectangle: "))
print("Rectangle Area:", rectangle.area(length, breadth))
print("Rectangle Perimeter:", rectangle.perimeter(length, breadth))

# Circle
radius = int(input("\nEnter radius of circle: "))
print("Circle Area:", circle.area(radius))
print("Circle Perimeter:", circle.perimeter(radius))

# Cuboid
```

```
length = int(input("\nEnter length of cuboid: "))
breadth = int(input("Enter breadth of cuboid: "))
height = int(input("Enter height of cuboid: "))
print("Cuboid Surface Area:", cuboid.Surface_area(length, breadth, height))
print("Cuboid Volume:", cuboid.Volume(length, breadth, height))

# Sphere
radius = int(input("\nEnter radius of sphere: "))
print("Sphere Surface Area:", sphere.surface_area(radius))
print("Sphere Volume:", sphere.volume(radius))
```

OUTPUT

```
Enter length of rectangle: 4
Enter breadth of rectangle: 3
Rectangle Area: 12
Rectangle Perimeter: 14

Enter radius of circle: 5
Circle Area: 78.53981633974483
Circle Perimeter: 31.41592653589793

Enter length of cuboid: 4
Enter breadth of cuboid: 5
Enter height of cuboid: 6
Cuboid Surface Area: 148
Cuboid Volume: 120

Enter radius of sphere: 3
Sphere Surface Area: 113.09733552923255
Sphere Volume: 113.09733552923254
```

2. Create a user-defined module to perform basic arithmetic operations. Import it into another program to perform arithmetic operations on two numbers given as input.

PROGRAM

addition.py

```
def add(a, b):  
    return a + b
```

subtraction.py

```
def sub(a, b):  
    return a - b
```

multiplication.py

```
def mul(a, b):  
    return a * b
```

division.py

```
def div(a, b):  
    if b == 0:  
        return "Error: Division by zero is not allowed."  
    return a / b
```

main.py

```
from ArithmeticOperations import addition, subtraction, multiplication, division  
  
num1 = int(input("Enter first number: "))  
num2 = int(input("Enter second number: "))  
  
#Addition  
print("Addition:", addition.add(num1, num2))  
  
#Subtraction  
print("Subtraction:", subtraction.sub(num1, num2))  
  
#Multiplicaion
```

```
print("Multiplication:", multiplication.mul(num1, num2))  
  
#Division  
  
print("Division:", division.div(num1, num2))
```

OUTPUT

```
Enter first number: 20  
Enter second number: 5  
Addition: 25  
Subtraction: 15  
Multiplication: 100  
Division: 4.0
```

COURSE OUTCOME 4

Date: 30-10-2025

Implement object oriented programming and exception handling.

1. Design a Python class Student with attributes name, roll number, marks and methods to display and update marks. Take user input to create a student object.

PROGRAM

```
class Student:
    def __init__(self, name,rolno,mrk):
        self.name=name    #self constructor is used to initilize all attribute
        self.rolno=rolno
        self.mrk=mrk

    def display(self):
        print("Name: ",self.name)
        print("Roll-No: ",self.rolno)
        print("Marks: ",self.mrk)

    def update(self,newmrk):
        self.mrk=newmrk

# Main
name=input("\nEnter student name: ")
rolno=input("Enter student rollno: ")
mrk=int(input("Enter student marks: "))
s1=Student(name,rolno,mrk)

#display the initial details
```

```
print("\n Student Details:")
s1.display()

#Update marks
newmrk=int(input("\nEnter newmarks to update: "))
s1.update(newmrk)

#display updated details
print("\n Updated Student Details:")
s1.display()
```

OUTPUT

```
Enter student name: Sandhra Snthose
Enter student rollno: MCA32
Enter student marks: 89
```

```
Student Details:
Name: Sandhra Snthose
Roll-No: MCA32
Marks: 89
```

```
Enter newmarks to update: 96
```

```
Updated Student Details:
Name: Sandhra Snthose
Roll-No: MCA32
Marks: 96
```

2. Create a class Student with attributes –Name and Marks of two subjects, each out of 100. Calculate the total marks and display it.

PROGRAM

```
class Student:
    def __init__(self, name, mrk1, mrk2):
        self.name=name
        self.mrk1=mrk1
        self.mrk2=mrk2
    def total(self):
        return self.mrk1+self.mrk2

    def display(self):
        print("\nName: ",self.name)
        print("Mark1:",self.mrk1)
        print("Mark2: ",self.mrk2)

#main
name=input("Enter the name of student: ")
mrk1=int(input("Enter the mark of 1st sub:"))
mrk2=int(input("Enter the mark of 2nd sub:"))

#object creation
s=Student(name,mrk1,mrk2)
print("Details of student:")
s.display()
print("Total mark: ",s.total())
```

OUTPUT

Enter the name of student: Hiba Ansar

Enter the mark of 1st sub:89

Enter the mark of 2nd sub:88

Details of student:

Name: Hiba Ansar

Mark1: 89

Mark2: 88

Total mark: 177

3. Create a Python class named Student with two data members representing marks in two subjects (each out of 100). Define a method to display the total marks of a student. Use operator overloading (+) to add the total marks of two Student objects and display the combined total. Accept marks as user input.

PROGRAM

```
class Student:
    def __init__(self,m1,m2):
        self.m1=m1
        self.m2=m2
    def total(self):
        return self.m1 + self.m2
    #operator overloading for '+'
    def __add__(self,other):
        total1=self.total()
        total2=other.total()
        return total1 + total2

#main prg
print("Enter the marks of 1st student")
s1m1=int(input("Enter the mark for 1st subject: "))
s1m2=int(input("Enter the mark for 2nd subject: "))
print("\nEnter the marks of 2nd student")
s2m1=int(input("Enter the mark for 1st subject: "))
s2m2=int(input("Enter the mark for 2nd subject: "))

#object for 1st student
s1=Student(s1m1,s1m2)
#object for 2nd student
s2=Student(s2m1,s2m2)
```

```
#calculate individual totals  
print("Total marks of student 1: ",s1.total())  
print("Total marks of student 2: ",s2.total())  
#using overloaded '+' operator  
ctotal=s1+s2  
print("Combined total of both Students: ",ctotal)
```

OUTPUT

```
Enter the marks of 1st student  
Enter the mark for 1st subject: 89  
Enter the mark for 2nd subject: 79  
Enter the marks of 2nd student  
Enter the mark for 1st subject: 99  
Enter the mark for 2nd subject: 96  
Total marks of student 1: 168  
Total marks of student 2: 195  
Combined total of both Students: 363
```

4. Create a class Student with attributes –Name and Marks of two subjects, each out of 100. Calculate the total marks and display it.

PROGRAM

```
class Rectangle:
    def __init__(self,l,b):
        self.l=l
        self.b=b

    def area(self):
        return self.l * self.b

    def __lt__(self,other):
        return self.area()<other.area()

#main program
print("Details of 1st Rectangele")
l1=int(input("Enter the length of 1st rectangle: "))
w1=int(input("Enter the width of 1st rectangle: "))

print("\nDetails of 2st Rectangele")
l2=int(input("Enter the length of 2st rectangle: "))
w2=int(input("Enter the width of 1st rectangle: "))

#object creation
r1=Rectangle(l1,w1)
r2=Rectangle(l2,w2)

if r1<r2 :
```

```
print("Area of Rectangle 1 is small ")  
else:  
    print("Area of Rectangle 2 is small")
```

OUTPUT

Details of 1st Rectangele

Enter the length of 1st rectangle: 3

Enter the width of 1st rectangle: 5

Details of 2st Rectangele

Enter the length of 2st rectangle: 4

Enter the width of 1st rectangle: 2

Area of Rectangle 1 is large

5. Create a Python program to demonstrate inheritance. Define a parent class Student that stores the name of a student and display it using a method. Define a child class MCA Student that inherits from Student and add an additional data member for the semester. Use the super() function to call the parent class constructor. Display the student's name and semester using appropriate methods. Accept the values through user input.

PROGRAM

```
class Student:
    def __init__(self,name):
        self.name=name

    def display(self):
        print("Name =",self.name)

# child class inheriting from parent class
class MCASStudent(Student):
    def __init__(self,name,sem):

        # call parent class constructor using super
        super().__init__(name) # using super to access name from parent
        self.sem=sem

    def showdetail(self):
        self.display()
        print("Semester = ",self.sem)

# MAIN
name=input("Enter the Student name: ")
sem=input("Enter the semester : ")

# create object of child class
```

```
s=MCASStudent(name,sem)
print("Student details are :")
s.showdetail()
```

OUTPUT

Enter the Student name: Hiba Ansar

Enter the semester : s1

Student details are :

Name = Hiba Ansar

Semester = s1

COURSE OUTCOME 5

Date: 05-11-2025

Manage files and form regular expressions for effective search operations on strings and files.

1. Write a program to create a file named student.txt and store student details such as roll number, name, and age.

PROGRAM :

Student.txt

```
Roll Number,name,age
MCA01,Hiba ANsar,21
MCA12,Sandhra Santhose,20
MCA31,Megha P,22
MCA21,Naiza C D,20
MCA13,MAiza Mansoor,18
```

Code:

```
f=open("student.txt")
print(f.read())
f.close()
```

OUTPUT

```
Roll Number,name,age
MCA01,Hiba ANsar,21
MCA12,Sandhra Santhose,20
MCA31,Megha P,22
MCA21,Naiza C D,20
MCA13,MAiza Mansoor,18
```

#

2. Write a program to read a text file line by line and store it into a list.

PROGRAM :

Textfile.txt

Good Morning
Hi Python
Better
Good Luck

Code:

```
with open("textfile.txt","r") as f:  
    line=f.read()  
list=line.splitlines()  
print("The text : ",line)  
print("The list : ",list)
```

OUTPUT

The text : Good Morning
Hi Python
Better
Good Luck

The list : ['Good Morning', 'Hi Python', 'Better ', 'Good Luck']

3. Write a program to read each row from a CSV file with fields roll number, name, age, course and display a list of strings.

PROGRAM :**Student.csv**

Roll-number	name	age	course
120	Hiba	21	MCA
142	megha	12	MBA
145	sandra	32	BBA
123	naiza	34	BCA
345	malu	36	BCom

Code:

```
import csv
with open("student.csv") as f:
    print(f.read())
    csvr=csv.reader(f)
    for row in csvr :
        print (row)
```

OUTPUT

```
Roll-number,name,age,course
120,hiba,21,MCA
142,megha,12,MBA
145,sandra,32,BBA
123,naiza,34,BCA
345,malu,36,BCom
```

4. Write a program to read specific columns of a CSV "comma separated values" file book with fields book number, title, author, price. Display specific columns, book number, title and price.

PROGRAM :**Book.csv**

book number	title	author	price
bk021	Tom And Jerry	Adheem JK	500
bk321	Harry Potter	Rowlling	850
bk435	Good of Evil Three	Aflal	800
bk523	Musquesters	Megha	900
bk657	fairy Tales	Hiba	450
tf564	hjpgfh	hgj	890

Code:

```
import csv
with open("book.csv",mode="r") as f:
    csvr=csv.reader(f)
    print("contents of file are: ")
    for row in csvr:
        print (row)
    f.close()
f=open("book.csv","r")
col=csv.reader(f)
print("The Specific Columes Are: ")
for i in col:
    print(i[0],i[1],i[2],i[3])
f.close()
```

OUTPUT

Contents of file are:

```
['book number', 'title', 'author', 'price']  
['bk021', 'Tom And Jerry', 'Adheem', '500']  
['bk321', 'Harry Potter', 'JK Rowling', '850']  
['bk435', 'Good of Evil', 'Aflal', '800']  
['bk523', 'Three Musquesters', 'Megha', '900']  
['bk657', 'fairy Tales', 'Hiba', '450 ']  
['tf564', 'hjgfjh', 'hgj', '890']
```

The Specific Columns Are:

```
book number title author price  
bk021 Tom And Jerry Adheem 500  
bk321 Harry Potter JK Rowling 850  
bk435 Good of Evil Aflal 800  
bk523 Three Musquesters Megha 900  
bk657 fairy Tales Hiba 450  
tf564 hjgfjh hgj 890
```

5. Using Regular Expression, search for a particular word (given as input) in a text file. Count the number of occurrences of the word in the file.

PROGRAM :**Word.txt**

Python is great. python is powerful. I love PYTHON!

Code:

```
import re
with open("word.txt","r") as f:
    txt=f.read()
    print(txt)
word=input("Enter the word to search: ")
occ=re.findall(word,txt,re.IGNORECASE)
count=len(occ)
print(occ)
print("The word ",word,"occures ",count, "times")
```

OUTPUT

Python is great. python is powerful. I love PYTHON!

Enter the word to search: is

['is', 'is']

The word is occures 2 times